ConstituencyIQ

AI Technology Stack & Architecture

COMPLETE AI LAYER DESIGN

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The Core Principle

ConstituencyIQ is not a single AI — it is an orchestra of 7 specialised AI systems working together. Each handles a specific layer of intelligence: reading, seeing, speaking, analysing, predicting, visualising, and conversing. No single AI does all of this well. The right architecture combines the best tool for each job.

1. AI Capability Map

The platform requires 7 distinct AI capabilities. Each is handled by a purpose-built AI system. Together they form a complete intelligence pipeline — from raw data in, to spoken conversation out.

#	AI Capability	What It Does	Best Tool	Phase
1	Data Reader	Reads, understands & extracts meaning from structured + unstructured data	Claude 3.5 Sonnet	1
2	Image / Photo Reader	Reads photos from field (voter ID, infrastructure, crowd, handwritten forms)	Claude Vision + Google Vision	1
3	Chart & Visual Generator	Converts data into charts, infographics, and ward-level visual reports	Claude + Chart.js + GPT-4o	1
4	Situation Analyser	Multi-variable analysis: ground + online data → insights & scoring	Claude 3.5 Sonnet (RAG)	1
5	Prediction Engine	Future vote share, swing analysis, scenario modelling	Claude + statistical model	2
6	IVRS Voice AI	Contextual, personalised phone conversations with voters	ElevenLabs + Sarvam AI + Claude	2
7	Multimodal Orchestrator	Combines all AI outputs into one unified intelligence layer	Claude API (system prompt + tools)	2

The sections below detail each AI system — what it does, which tool to use, why, and exactly how it fits into ConstituencyIQ.

2. AI #1 — Data Reader

READS AND UNDERSTANDS EVERY TYPE OF DATA THE PLATFORM RECEIVES

The Data Reader is the foundation of the entire intelligence stack. Before any analysis, prediction, or recommendation can happen, the AI must first understand what the data says — not just count it, but comprehend it.

2.1 What It Needs to Read

Data Type	Source	Challenge
Structured survey data	KoboToolbox / SurveyMonkey	Inconsistent field names, missing values
Excel / CSV uploads	Campaign manager uploads	Mixed formats, merged cells, Hindi text
Voter rolls (PDF)	ECI website	Scanned PDFs, tabular data in image format
Local news articles	Web scraping	Unstructured text, regional languages
Social media posts	Facebook, Twitter/X	Short text, slang, Hinglish, sarcasm
Handwritten field notes	Photo from field worker	OCR + comprehension required
Government scheme data	MGNREGA, PM Awas portals	Semi-structured tables, state-specific formats
Census data	Census 2011/2021 PDFs	Dense tables, ward-level hierarchies

2.2 Recommended Tool: Claude 3.5 Sonnet (Anthropic)

🧠 **Claude 3.5 Sonnet** | *Primary Data Reader & Intelligence Engine*

Tool: Anthropic API | **Cost:** ~$3 per million tokens (input)

Capabilities:

- Reads structured JSON, CSV, and table data natively — understands column headers, data types, and relationships
- Comprehends unstructured text in English, Hindi, and Hinglish without any translation step
- Extracts key entities: names, places, issues, sentiment, political affiliations from raw text
- Identifies anomalies and data quality issues (duplicates, outliers, suspicious patterns)
- Synthesises across multiple data sources simultaneously in a single prompt
- 200,000 token context window — can read an entire constituency's survey data in one call
- Produces structured JSON output for downstream processing (chart generation, scoring)

2.3 Supporting Tool: Google Document AI

For scanned PDFs and official government documents (voter rolls, census tables) that are image-based:

Google Document AI — Scanned Document Reader

- Extracts tabular data from scanned ECI voter roll PDFs with high accuracy
- Handles Hindi and regional language text in official government documents

- Returns structured JSON output that Claude then analyses and interprets
- Cost: ~$1.50 per 1,000 pages — affordable for constituency-scale documents
- Integration: Simple REST API, output plugs directly into Claude's analysis pipeline

2.4 How It Works in Practice

Field worker submits KoboToolbox survey → Data Reader pipeline:

1. KoboToolbox webhook fires → raw JSON lands in ConstituencyIQ backend
2. Claude normalises the data: standardises field names, fills inferred values, detects language
3. Claude extracts key signals: primary issue, voter leaning, community, economic status
4. Enriched record stored in Firestore with AI-generated tags and sentiment scores
5. Popularity Score and Action Engine automatically update with new data point

3. AI #2 — Image & Photo Reader

READS PHOTOS, HANDWRITTEN FORMS, INFRASTRUCTURE IMAGES, AND VOTER ID CARDS

Field workers collect far more than typed data. They photograph broken roads, damaged infrastructure, community events, handwritten notes, and voter ID cards. The Image Reader AI extracts actionable intelligence from all of these.

3.1 Photo Types the Platform Will Receive

Photo Type	What AI Extracts	Use Case
Voter ID card	Name, address, voter ID number, booth number	Auto-fill voter record, deduplication
Infrastructure (road, drain, school)	Condition assessment, location, severity rating	Evidence for micro-action recommendations
Handwritten survey form	All field data from written form	Offline data entry by low-literacy workers
Crowd / rally photo	Estimated crowd size, crowd demographics, energy level	Competitor intelligence, own event assessment
Market / locality photo	Business type count, infrastructure quality, footfall estimate	Micro-action targeting (vendor zone, road repair)
Competitor poster / banner	Candidate name, party, key slogans, promises made	Competitor analysis module
News clipping photo	Full article text + publication + date	Local news intelligence

3.2 Recommended Tool: Claude Vision (Primary)

👁️ Claude Vision | *Multimodal Image Understanding*

Tool: Anthropic API (same as Data Reader) | **Cost:** Included in Claude API pricing

Capabilities:

- Reads and understands image content with full contextual reasoning — not just labelling objects
- Extracts text from images (OCR) including Hindi, handwritten text, and low-quality scans
- Assesses infrastructure condition from photos: 'This road has severe potholes, estimated repair urgency: HIGH'
- Reads competitor banners and extracts promises, party details, and slogans
- Estimates crowd size from rally/event photos with confidence ranges
- Understands scene context: a photo of a broken drain near a school is higher priority than one in an open field
- Same API as the Data Reader — no additional integration required, simplifying the stack

3.3 Supporting Tool: Google Vision API (For Voter ID / Document OCR)

Google Vision API — Structured Document Extraction

- Best-in-class for extracting structured data from ID cards (Voter ID, Aadhaar format)
- Extremely fast — sub-second response for a single card photo
- Returns bounding boxes so UI can show what was extracted and where on the card
- Handles blurry, angled, or partially lit ID photos far better than Claude Vision
- Cost: $1.50 per 1,000 image units — very affordable at field survey scale
- Integrates with Claude: Google Vision extracts raw text → Claude parses and maps to voter record fields

3.4 Field Worker Photo Workflow

How a field worker submits a photo and it becomes intelligence:

6. Field worker opens Android app → taps 'Attach Photo' → selects category (Voter ID / Infrastructure / Other)
7. Photo uploads to Firebase Storage → triggers Cloud Function
8. If Voter ID: Google Vision API extracts text → Claude maps to voter schema → auto-fills form
9. If Infrastructure: Claude Vision analyses condition, assigns severity score, extracts location context
10. If Handwritten form: Claude Vision reads all fields → populates digital form for worker to verify
11. Result stored in Firestore with image URL, AI analysis, confidence score, and timestamp
12. Campaign Manager sees photo + AI summary in dashboard — one tap to approve or flag

4. AI #3 — Chart & Visual Generator

CONVERTS RAW DATA INTO CHARTS, INFOGRAPHICS, MAPS, AND VISUAL REPORTS

Data alone is not enough for a candidate or campaign manager. They need to see the information — instantly, clearly, and without needing to read a table. The Visual Generator converts any dataset or AI analysis into a visual automatically.

4.1 Visuals the Platform Must Generate

Visual Type	Data Source	Tool
Popularity Score gauge (0–100)	Weighted formula + Claude	Chart.js / D3.js (frontend)
Ward-by-ward sentiment heatmap	Ground survey + Claude scoring	Google Maps API + heat layer
Voter segment donut chart	KoboToolbox / survey data	Chart.js (auto-generated)
Issue priority bar chart	Survey field: Problem	Chart.js (auto-generated)
Vote share prediction timeline	Prediction engine output	Chart.js line chart
Micro-action impact bars	Action recommendation engine	Custom Chart.js
Competitor comparison radar	Competitor analysis module	Chart.js radar chart
AI-generated infographic PDF	Claude + ReportLab / Puppeteer	PDF export for sharing
Booth-level turnout map	ECI historical data	Google Maps + marker clusters

4.2 AI Role in Visual Generation

Most charts are generated automatically by the frontend using Chart.js. But AI plays three critical roles in the visual layer:

Claude: Decides WHAT to show

- Analyses the data and decides which chart type best represents it
- Generates the chart data structure (labels, values, colours) as JSON
- Writes the human-readable caption that appears below each chart
- Identifies the most important visual to show first based on what's changed
- Generates the narrative: 'This chart shows your support has dropped 12 points in Ward 5 since last week'

Chart.js / D3: Renders HOW it looks

- Renders bar, pie, donut, radar, and line charts from JSON data
- Google Maps API renders geographic heatmaps and booth markers
- Puppeteer (Node.js) converts any chart to PDF for sharing
- All charts are interactive — hover, click, drill down to ward level
- Colour coding is semantic: red = weak, green = strong, amber = swing

4.3 AI-Generated Infographic Reports

📊 GPT-4o (OpenAI) | *Visual Infographic & Report Generator*

Tool: OpenAI API | **Cost:** ~$5 per million tokens (input)

Capabilities:

- Generates publication-ready infographic layouts from structured data
- Creates visual 'Daily Briefing' for the candidate: one page, all key numbers, shareable on WhatsApp
- Produces ward-level summary cards: photo of locality + AI assessment + key stats in one visual
- Generates social media-ready graphics from data (for candidate's own Instagram/Facebook)
- Why GPT-4o specifically: its structured output with image generation (DALL-E 3) enables combined text+visual reports that Claude cannot produce natively
- Integration: Claude does the analysis → structured JSON → GPT-4o generates visual → stored in Firebase

5. AI #4 — Situation Analyser

THE CORE INTELLIGENCE ENGINE — SYNTHESISES EVERYTHING INTO ACTIONABLE INSIGHT

The Situation Analyser is the most sophisticated AI in the stack. It takes all incoming data — ground surveys, images, news, social media, historical results, census data — and synthesises them into a coherent picture of what is happening in the constituency right now.

5.1 What It Analyses

Multivariate Analysis Capabilities

- Caste × Ward × Issue × Vote History → identifies which sub-segments are genuinely persuadable
- Age × Gender × Occupation × Problem → surfaces non-obvious voter behaviour patterns
- Online sentiment × Ground survey → detects when media narrative diverges from ground reality
- Development gap × Community type → predicts which infrastructure project wins most votes per rupee
- Competitor activity × Ground sentiment shift → measures if opponent actions are moving numbers
- Time-series: surveys from Week 1 vs Week 3 → detects momentum — are things improving or declining?
- Geographic clustering: which wards are contiguous, similar, and should be treated as one campaign unit

5.2 Recommended Tool: Claude 3.5 Sonnet with RAG

🔍 Claude + RAG (Retrieval Augmented Generation) | *Situation Analysis & Constituency Intelligence*

Tool: Anthropic API + Pinecone (vector DB) | **Cost:** ~$20–50/month at pilot scale

Capabilities:

- RAG allows Claude to query a knowledge base of constituency-specific documents before answering
- Knowledge base contains: past news articles, ECI PDFs, census tables, government scheme data, all survey history
- Every question asked in the AI chat is answered using both real-time data AND stored constituency knowledge
- Produces the Popularity Score by weighing 5 data inputs with configurable weights per constituency type
- Generates the Micro-Action recommendations: for each ward, ranks possible actions by vote ROI
- Identifies the Expectation Index: what voters in each community expect, derived from open-text survey responses
- Runs weekly Situation Reports automatically — no human prompt needed, scheduled via cron job

5.3 RAG Architecture

RAG (Retrieval Augmented Generation) is the technique that gives Claude a long-term memory of the constituency:

13. All constituency documents are converted to vector embeddings using text-embedding-3-small (OpenAI)
14. Embeddings stored in Pinecone vector database (free tier: 1M vectors — sufficient for pilot)
15. When a question is asked, the most relevant documents are retrieved from Pinecone
16. Retrieved documents + the question are sent to Claude as context
17. Claude answers using both the retrieved knowledge AND the real-time survey data

18. This means Claude knows: what happened in 2019 election in this constituency, which roads are broken, what the local newspaper said last week — all without re-reading every document every time

6. AI #5 — Prediction Engine

VOTE SHARE PREDICTION, SWING ANALYSIS, AND SCENARIO MODELLING

The Prediction Engine combines statistical modelling with AI reasoning to forecast election outcomes. It is updated continuously as new data arrives — giving the campaign a live probability of winning.

6.1 Prediction Model Inputs

Input	Weight	Source
Voter support sentiment (direct)	35%	KoboToolbox field survey
Historical baseline (past 3 elections)	20%	ECI open data
Community alignment score	20%	Survey × demographic model
Online sentiment trend	15%	News + social media scraping
Booth-level turnout prediction	10%	Historical ECI + field observation

6.2 Recommended Approach: Claude + Python Statistical Model

📈 Claude + Python (scikit-learn) | *Hybrid AI + Statistical Prediction*

Tool: Anthropic API + Python backend | **Cost:** Covered by existing API costs

Capabilities:

- Python runs a weighted regression model using the 5 inputs above → outputs raw vote share estimate
- Claude then contextualises the number: 'This 38% estimate assumes similar turnout to 2019. If turnout drops in Ward 5 (historically low enthusiasm), the range narrows to 36–40%'
- Scenario modelling: candidate or campaign manager inputs a hypothetical action → Claude estimates the vote impact
- Example: 'If we complete the Ward 3 road repair, predicted vote share increases from 38% to 41% based on similar infrastructure-sentiment correlations in past surveys'
- Confidence intervals widen when data is sparse and tighten as more surveys come in
- Booth-level prediction: colours each booth on the map — won, losing, swing — with estimated margin

7. AI #6 — IVRS Voice AI (The Most Complex Feature)

CONTEXTUAL, PERSONALISED PHONE CONVERSATIONS WITH VOTERS AT SCALE

This is the most technically sophisticated and highest-value feature in the platform. The IVRS AI calls voters and holds a real, contextual conversation — not a recorded message, not a simple script, but a dynamic conversation that adapts based on who the voter is, what they care about, and what they say.

7.1 What Makes This Different from Normal IVRS

❌ Traditional IVRS (What exists today)

- 'Press 1 for Hindi, Press 2 for English'
- Same recorded message to every voter
- Cannot handle any question or deviation
- Zero personalisation — doesn't know who it's calling
- Voters hang up within 10 seconds
- No contextual awareness whatsoever

✅ ConstituencyIQ Voice AI (What we build)

- Speaks in the voter's language, dialect, and tone automatically
- Knows the voter's name, ward, primary issue, community, family situation
- Says: 'Ramesh ji, aapke Ward 3 mein jo road repair ka issue tha, uske baare mein...'
- Listens and responds to what the voter actually says
- Adapts conversation based on voter's sentiment in real time
- Records voter responses back into the database as new data

7.2 The Full AI Stack for IVRS

IVRS requires four AI components working in sequence, in under 1 second per exchange:

🎙️ Sarvam AI (STT) | *Speech-to-Text — Voter's voice → Text*

Tool: Sarvam AI API (Indian startup) | **Cost:** ~₹0.5 per minute of audio

Capabilities:

- Best-in-class speech recognition for Indian languages — Hindi, Hinglish, Bhojpuri, Marathi, Tamil
- Handles accents, background noise (market, street), and low-quality phone audio
- Sub-200ms transcription — fast enough for real conversation
- Understands 'haan', 'nahi', 'sahi hai', 'kya bolein', 'ek second' — natural Indian phone conversation patterns
- Why Sarvam over Google/AWS: trained specifically on Indian languages and phone audio, far higher accuracy
- Provides word-level confidence scores — low confidence = AI asks for clarification

🧠 Claude 3.5 Haiku | *Conversation Brain — Decides what to say next*

Tool: Anthropic API | **Cost:** ~$0.25 per million tokens (fastest + cheapest Claude)

Capabilities:

- Receives: voter profile from database (name, ward, caste, primary issue, past vote, family size, economic status)
- Receives: full conversation history so far
- Receives: what the voter just said (from Sarvam STT)
- Produces: the exact next response — personalised, contextual, natural

- Uses Haiku (not Sonnet) for speed — response must come in under 500ms for natural conversation flow
- System prompt contains the voter's complete situation analysis: their issues, their likely concerns, objections they may raise, and what the candidate has done or promised for their specific area
- Can handle: voter complaint, voter support, voter asking about rival candidate, voter asking about a specific issue, voter being rude or hanging up
- Updates voter record after call: noted support level, key concern raised, call outcome

🔊 ElevenLabs (TTS) | *Text-to-Speech — AI response → Natural Voice*

Tool: ElevenLabs API | **Cost:** ~$0.30 per 1,000 characters

Capabilities:

- Converts Claude's text response into a natural, warm, human-sounding Hindi/Hinglish voice
- Multiple voice options — choose a voice that matches the candidate's brand (warm, authoritative, friendly)
- Emotion-aware: can speak with empathy when voter raises a problem, with confidence when discussing achievements
- Sub-300ms generation — fast enough for conversation
- Custom voice cloning option (Phase 3): clone the candidate's own voice so the AI literally speaks as the candidate
- Handles code-switching: seamlessly moves between Hindi and English mid-sentence

📞 Exotel / Tata Tele (Telephony) | *Call Infrastructure — Connects AI to voter's phone*

Tool: Exotel API (Indian cloud telephony) | **Cost:** ~₹0.35 per minute of call

Capabilities:

- Indian cloud telephony provider — handles bulk outbound calling at scale
- Provides WebSocket connection that streams audio in real time to Sarvam + ElevenLabs
- Handles call routing, retry logic, do-not-disturb compliance
- Provides call logs, recording, and analytics natively
- Why Exotel over Twilio: Indian DID numbers, better Indian carrier connectivity, lower latency, INR billing
- Supports 10,000+ simultaneous calls — enough for a full Vidhan Sabha constituency outreach
- Compliant with TRAI regulations for political calls

7.3 The IVRS Conversation Flow

Here is an example of a real conversation the AI would have — personalised to one specific voter:

Speaker	Conversation
AI (calls)	Namaste, kya mein Ramesh Kumar ji se baat kar sakta hoon? Main Rajesh Sharma ji ke campaign ki taraf se call kar raha hoon.
Voter	Haan, main Ramesh bol raha hoon. Kaun hai?
AI	Ramesh ji, main Rajesh Sharma ji ka sahayak hoon. Aap Ward 3, Bazaar gali mein rehte hain — sahi hai na? Aapke area mein road ka jo muddha hai, uske baare mein Rajesh ji ne kuch kaam shuru kiya hai.
Voter	Haan, road toh bahut kharab hai. Kab thik hogi?

AI	Bilkul samajh sakta hoon aapki problem. Rajesh ji ne Ward 3 ki road repair ke liye PWD ke saath baat ki hai — 15 tarikh tak kaam shuru hoga. Aur aapke area ke vendors ke liye ek permanent market jagah ka proposal bhi hai. Aap is baare mein kya sochte hain?
Voter	Yeh toh achha hoga. Market road banne se humara dhanda bhi badega.
AI (records internally)	[Voter: Supportive. Primary issue: Road + Market. Sentiment: Positive. Follow-up: Send WhatsApp message about market road proposal. Call outcome: STRONG_SUPPORT]

Every word in this conversation was dynamically generated by Claude, drawing from: Ramesh's survey record, his ward's issues, the candidate's actual promises, and the conversation history.

8. Complete AI Stack & Cost Estimate

Summary of all AI tools, their role, phase, and estimated monthly cost at pilot scale (one Vidhan Sabha constituency, ~500 surveys, ~2,000 IVRS calls/month).

AI System	Tool	Role	Phase	Est. Cost/mo
Data Reader	Claude 3.5 Sonnet	Analysis, insights, chat	Phase 1	₹800–2,000
Image Reader	Claude Vision	Photo analysis, scene understanding	Phase 1	₹300–800
Document OCR	Google Vision API	Voter ID, PDF extraction	Phase 1	₹200–500
Visual Generator	Chart.js + GPT-4o	Charts, infographics	Phase 1	₹500–1,500
Situation Analyser	Claude + Pinecone RAG	Deep constituency intelligence	Phase 1–2	₹400–1,000
Prediction Engine	Claude + Python	Vote share, scenarios	Phase 2	₹200–600
IVRS Voice STT	Sarvam AI	Voice → text (Indian languages)	Phase 2	₹1,500–5,000
IVRS Brain	Claude 3.5 Haiku	Conversation reasoning	Phase 2	₹400–1,200
IVRS Voice TTS	ElevenLabs	Text → natural voice	Phase 2	₹1,000–3,000
Telephony	Exotel	Phone call infrastructure	Phase 2	₹2,000–8,000
Vector DB	Pinecone (free tier)	RAG knowledge store	Phase 1–2	₹0 (free tier)
Embeddings	OpenAI text-embedding-3	Document vectorisation	Phase 1–2	₹100–300
TOTAL (Phase 1)		**Dashboard + chat + images + charts**		**₹2,200–5,800/mo**
TOTAL (Phase 2)		**+ IVRS + prediction + RAG**		**₹7,000–22,000/mo**

At a Pro tier pricing of ₹75,000/election cycle, even Phase 2 AI costs are easily absorbed with strong margins.

8.1 The Golden Rule of This AI Stack

One Principle: Right AI for the Right Job

- Claude = Reasoning, analysis, conversation, reading complex data, writing responses
- Sarvam AI = Indian language speech recognition (Claude cannot do this)
- ElevenLabs = Natural voice synthesis in Hindi/Hinglish (Claude cannot do this)
- Google Vision = Fast, structured OCR for ID cards and official documents
- GPT-4o = Visual infographic generation where Claude's text output is combined with image creation
- Pinecone = Long-term memory — stores constituency knowledge so Claude doesn't start fresh each time
- Exotel = Phone call infrastructure — this is not AI, it is the pipe the AI speaks through

- DO NOT try to make one AI do everything — latency, cost, and accuracy will all suffer

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